

AZAD NAEEMI

professor

基本信息

姓名: AZAD NAEEMI

所属院校: 佐治亚理工学院

邮箱: azad@gatech.edu

地区: United States

学科领域

微电子科学与工程

集成电路设计与集成系统

应用物理学

材料物理

研究领域

微电子器件与制造

纳米电子与先进工艺

超大规模集成电路设计

数字集成电路设计

低功耗电路与能量管理

EDA工具与电路验证

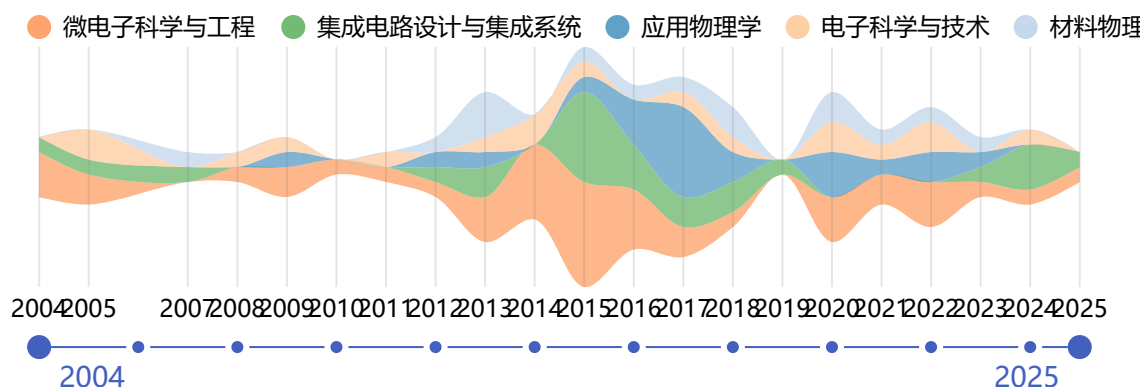
磁学与自旋电子学

磁性材料与磁学

个人简介

AZAD NAEEMI is a Professor in the School of Electrical and Computer Engineering at the Georgia Institute of Technology. He holds degrees from Sharif University and Georgia Tech. Naeemi's research focuses on integrated circuits using conventional and emerging nanoelectronic and spintronic devices. He has received the IEEE EDS Paul Rappaport Award, the IEEE Solid-State Circuits Society James D. Meindl Innovators Award, an NSF CAREER Award, and several best paper awards. His work spans materials, devices, circuits, and systems. Contact: azad@gatech.edu, Phone: 404.894.4829, Office: Pettit/MiRC 216.

近年发表的论文



Cross-Layer Modeling and Design of Content Addressable Memories in Advanced Technology Nodes for Similarity Search

IEEE TRANSACTIONS ON ELECTRON DEVICES

Narla, Siri; Kumar, Piyush; Adnaan, Mohammad; Naeemi, Azad

被引用次数: 0

2025

Impact of Technology Scaling and Back-End-of-the-Line Technology Solutions on Magnetic Random-Access Memories

IEEE JOURNAL ON EXPLORATORY SOLID-STATE COMPUTATIONAL DEVICES AND CIRCUITS

Kumar, Piyush; Shim, Da Eun; Narla, Siri; Naeemi, Azad

被引用次数: 1

2024

Compact Physical Model for Ferroelectric/Antiferroelectric/Dielectric Mixed Phase Capacitors

IEEE ELECTRON DEVICE LETTERS

Adnaan, Mohammad; Islam, Muhammad Mainul; Chang, Sou-Chi; Li, Hai; Young, Ian A.; Naeemi, Azad

被引用次数: 1

2024

Evaluation of a Plasmon-Based Optical Integrated Circuit for Error-Tolerant Streaming Applications

IEEE JOURNAL ON EXPLORATORY SOLID-STATE COMPUTATIONAL DEVICES AND CIRCUITS

Noor, Samantha Lubaba; Wu, Xuan; Lin, Dennis; van Dorpe, Pol; Catthoor, Francky; Reynaert, Patrick; Naeemi, Azad

被引用次数: 0

2024

Comparison of Photonic to Plasmonic Mode Converters for Plasmonic Multiple-Input Devices

IEEE PHOTONICS JOURNAL

Noor, Samantha Lubaba; Catthoor, Francky; Lin, Dennis; Van Dorpe, Pol; Naeemi, Azad

被引用次数: 1

2024

A Compact Model for Ferroelectric Capacitors Based on Multidomain Phase-Field Framework

IEEE TRANSACTIONS ON ELECTRON DEVICES

Adnaan, Mohammad; Chang, Sou-Chi; Li, Hai; Nikonov, Dmitri; Young, Ian A.; Naeemi, Azad

被引用次数: 2

2023

A Comprehensive Modeling Platform for Interconnect Technologies

IEEE TRANSACTIONS ON ELECTRON DEVICES

Shim, Da Eun; Huang, Victor; Chen, Xinkang; Gupta, Sumeet K.; Naeemi, Azad

被引用次数: 3

2023

Design of a Compact Spin-Orbit-Torque-Based Ternary Content Addressable Memory

IEEE TRANSACTIONS ON ELECTRON DEVICES

Narla, Siri; Kumar, Piyush; Laguna, Ann Franchesca; Reis, Dayane; Hu, X. Sharon; Niemier, Michael; Naeemi, Azad

被引用次数: 7

2023

A Drift-Diffusion Based Modeling and Optimization Framework for Nanoscale Spin-Orbit Torque Devices

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kumar, Piyush; Liao, Yu-Ching; Ralph, Daniel C.; Naeemi, Azad

被引用次数: 1

2023

Towards Area Efficient Logic Circuit: Exploring Potential of Reconfigurable Gate by Generic Exact Synthesis

IEEE OPEN JOURNAL OF THE COMPUTER SOCIETY

Shang, Liuting; Naeemi, Azad; Pan, Chenyun

被引用次数: 0

2023

Quantum games and interactive tools for quantum technologies outreach and education

OPTICAL ENGINEERING

Seskir, Zeki C.; Migdal, Piotr; Weidner, Carrie; Anupam, Aditya; Case, Nicky; Davis, Noah; Decaroli, Chiara; Ercan, Ilke; Foti, Caterina; Gora, Pawel; Jankiewicz, Klementyna; La Cour, Brian R.; Malo, Jorge Yago; Maniscalco, Sabrina; Naeemi, Azad; Nita, Laurentiu; Parvin, Nassim; Scafrimuto, Fabio; Sherson, Jacob F.; Surer, Elif; Wootton, James; Yeh, Lia; Zabello, Olga; Chiofalo, Marliu

被引用次数: 22

2022

Evaluating the Performances of the Ultralow Power Magnetoelectric Random Access Memory With a Physics-Based Compact Model of the Antiferromagnet/Ferromagnet Bilayer

IEEE TRANSACTIONS ON ELECTRON DEVICES

Liao, Yu-Ching; Hsu, Chia-Sheng; Nikonov, Dmitri; Chang, Sou-Chi; Li, Hai; Young, Ian A.; Naeemi, Azad

被引用次数: 7

2022

Plasmonic MIM and MSM Waveguide Couplers for Plasmonic Integrated Computing System

IEEE PHOTONICS JOURNAL

Noor, Samantha Lubaba; Van Dorpe, Pol; Lin, Dennis; Catthoor, Francky; Naeemi, Azad

被引用次数: 3

2022

Bipolar Electric-Field Switching of Perpendicular Magnetic Tunnel Junctions through Voltage-Controlled Exchange Coupling

NANO LETTERS

Zhang, Delin; Bapna, Mukund; Jiang, Wei; Sousa, Duarte; Liao, Yu-Ching; Zhao, Zhengyang; Lv, Yang; Sahu, Prottyush; Lyu, Deyuan; Naeemi, Azad; Low, Tony; Majetich, Sara A.; Wang, Jian-Ping

被引用次数: 26

2022

Modeling and Design for Magnetoelectric Ternary Content Addressable Memory (TCAM)

IEEE JOURNAL ON EXPLORATORY SOLID-STATE COMPUTATIONAL DEVICES AND CIRCUITS

Narla, Siri; Kumar, Piyush; Laguna, Ann Franchesca; Reis, Dayane; Sharon, X. Sharon; Niemier, Michael; Naeemi, Azad

被引用次数: 10

2022

Physics-Based Models for Magneto-Electric Spin-Orbit Logic Circuits

IEEE JOURNAL ON EXPLORATORY SOLID-STATE COMPUTATIONAL DEVICES AND CIRCUITS

Li, Hai; Nikonov, Dmitri E.; Lin, Chia-Ching; Camsari, Kerem; Liao, Yu-Ching; Hsu, Chia-Sheng; Naeemi, Azad; Young, Ian A.

被引用次数: 9

2022

Benchmarking of spin-orbit torque vs spin-transfer torque devices

APPLIED PHYSICS LETTERS

Kumar, Piyush; Naeemi, Azad

被引用次数: 9

2022

Nonvolatile Voltage Controlled Molecular Spin-State Switching for Memory Applications

MAGNETOCHEMISTRY

Ekanayaka, Thilini K.; Hao, Guanhua; Mosey, Aaron; Dale, Ashley S.; Jiang, Xuanyuan; Yost, Andrew J.; Sapkota, Keshab R.; Wang, George T.; Zhang, Jian; N'Diaye, Alpha T.; Marshall, Andrew; Cheng, Ruihua; Naeemi, Azad; Xu, Xiaoshan; Dowben, Peter A.

被引用次数: 35

2021

The 2021 Magnonics Roadmap

JOURNAL OF PHYSICS-CONDENSED MATTER

Barman, Anjan; Gubbiotti, Gianluca; Ladak, S.; Adeyeye, A. O.; Krawczyk, M.; Grafe, J.; Adelman, C.; Cotozana, S.; Naeemi, A.; Vasyuchka, V.; Hillebrands, B.; Nikitov, S. A.; Yu, H.; Grundler, D.; Sadovnikov, A.; V.; Grachev, A. A.; Sheshukova, S. E.; Duquesne, J-Y; Marangolo, M.; Csaba, G.; Porod, W.; Demidov, V. E.; Urazhdin, S.; Demokritov, S. O.; Albisetti, E.; Petti, D.; Bertacco, R.; Schultheiss, H.; Kruglyak, V. V.; Poimanov, V. D.; Sahoo, S.; Sinha, J.; Yang, H.; Munzenburg, M.; Moriyama, T.; Mizukami, S.; Landeros, P.; Gallardo, R. A.; Carlotti, G.; Kim, J.-V.; Stamps, R. L.; Camley, R. E.; Rana, B.; Otani, Y.; Yu, W.; Yu, T.; Bauer, G. E. W.; Back, C.; Uhrig, G. S.; Dobrovolskiy, O.; Budinska, B.; Qin, H.; van Dijken, S.; Chumak, A.; Khitun, A.; Nikonov, D. E.; Young, I. A.; Zingsem, B. W.; Winklhofer, M.

被引用次数: 415

2021

Impact of carrier mobility and lifetime on the potential performance of a plasmonic detector

2021 ANNUAL CONFERENCE OF THE IEEE PHOTONICS SOCIETY (IPC)

Noor, Samantha Lubaba; Srinivasan, Ashwyn; Haffner, Christian; De Greve, Kristiaan; Van Dorpe, Pol; Lin, Dennis; Catthoor, Francky; Naeemi, Azad

被引用次数: 0

2021

Hysteresis-Free Negative Capacitance Effect in Metal-Ferroelectric-Insulator-Metal Capacitors with Dielectric Leakage and Interfacial Trapped Charges

PHYSICAL REVIEW APPLIED

Hsu, Chia-Sheng; Chang, Sou-Chi; Nikonov, Dmitri E.; Young, Ian A.; Naeemi, Azad

被引用次数: 8

2021

Valley-Spin Logic Gates

PHYSICAL REVIEW APPLIED

Tao, L. L.; Naeemi, Azad; Tsymbal, Evgeny Y.

被引用次数: 32

2020

Materials Requirements of High-Speed and Low-Power Spin-Orbit-Torque Magnetic Random-Access Memory

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Li, Xiang; Lin, Shy-Jay; Dc, Mahendra; Liao, Yu-Ching; Yao, Chengyang; Naeemi, Azad; Tsai, Wilman; Wang, Shan X.

被引用次数: 21

2020

Accurate Determination of Interlayer Resistivity of 2-D Layered Systems: Graphene Case Study

IEEE TRANSACTIONS ON ELECTRON DEVICES

Nashed, Ramy; Pan, Chenyun; Wu, Xiangyu; Asselberghs, Inge; Tokei, Zsolt; Catthoor, Francky; Naeemi, Azad

被引用次数: 2

2020

A Theoretical Study of Multidomain Ferroelectric Switching Dynamics With a Physics-Based SPICE Circuit Model for Phase-Field Simulations

IEEE TRANSACTIONS ON ELECTRON DEVICES

Hsu, Chia-Sheng; Chang, Sou-Chi; Nikonov, Dmitri E.; Young, Ian A.; Naeemi, Azad

被引用次数: 11

2020

Modeling and Optimization of Plasmonic Detectors for Beyond-CMOS Plasmonic Majority Logic Gates

JOURNAL OF LIGHTWAVE TECHNOLOGY

Noor, Samantha Lubaba; Dens, Kristof; Reynaert, Patrick; Catthoor, Francky; Lin, Dennis; Van Dorpe, Pol; Naeemi, Azad

被引用次数: 5

2020

Understanding the Switching Mechanisms of the Antiferromagnet/Ferromagnet Heterojunction

NANO LETTERS

Liao, Yu-Ching; Nikonov, Dmitri E.; Dutta, Sourav; Chang, Sou-Chi; Hsu, Chia-Sheng; Young, Ian A.; Naeemi, Azad

被引用次数: 13

2020

A Mixed Signal Architecture for Convolutional Neural Networks

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Lou, Qiuwen; Pan, Chenyun; McGuinness, John; Horvath, Andras; Naeemi, Azad; Niemier, Michael; Hu, X. Sharon

被引用次数: 20

2019

Analytic modeling of dipolar field requirements for robust coupling in a non-identical biaxial two-magnet system

JOURNAL OF APPLIED PHYSICS

Kani, Nickvash; Rakheja, Shaloo; Naeemi, Azad

被引用次数: 1

2018

Clocked Magnetostriction-Assisted Spintronic Device Design and Simulation

IEEE TRANSACTIONS ON ELECTRON DEVICES

Iraei, Rouhollah Mousavi; Kani, Nickvash; Dutta, Sourav; Nikonov, Dmitri E.; Manipatruni, Sasikanth; Young, Ian A.; Heron, John T.; Naeemi, Azad

被引用次数: 5

2018

Field emission from graphene sheets and its application in floating gate memories

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Nashed, R.; Pan, C.; Brenner, K.; Naeemi, A.

被引用次数: 3

2018

Particle in a Box: An Experiential Environment for Learning Introductory Quantum Mechanics

IEEE TRANSACTIONS ON EDUCATION

Anupam, Aditya; Gupta, Ridhima; Naeemi, Azad; JafariNaimi, Nassim

被引用次数: 20

2018

Transient Performance Analysis and Optimization of Crossbar Memory Arrays Using NbO₂-Based Threshold Switching Selectors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Pan, Chenyun; Naeemi, Azad

被引用次数: 3

2018

Performance Analysis and Enhancement of Negative Capacitance Logic Devices Based on Internally Resistive Ferroelectrics

IEEE ELECTRON DEVICE LETTERS

Hsu, Chia-Sheng; Pan, Chenyun; Naeemi, Azad

被引用次数: 10

2018

Theoretical Approach to Electroresistance in Ferroelectric Tunnel Junctions

PHYSICAL REVIEW APPLIED

Chang, Sou-Chi; Naeemi, Azad; Nikonov, Dmitri E.; Gruverman, Alexei

被引用次数: 29

2017

Strain-Mediated Magnetization Reversal Through Spin-Transfer Torque

IEEE TRANSACTIONS ON MAGNETICS

Kani, Nickvash; Heron, John T.; Naeemi, Azad

被引用次数: 6

2017

Modeling Interconnect Variability at Advanced Technology Nodes and Potential Solutions

IEEE TRANSACTIONS ON ELECTRON DEVICES

Prasad, Divya; Pan, Chenyun; Naeemi, Azad

被引用次数: 8

2017

Interconnect Design and Benchmarking for Charge-Based Beyond-CMOS Device Proposals (vol 37, pg 508, 2016)

IEEE ELECTRON DEVICE LETTERS

Pan, Chenyun; Naeemi, Azad

被引用次数: 0

2017

Proposal for nanoscale cascaded plasmonic majority gates for non-Boolean computation

SCIENTIFIC REPORTS

Dutta, Sourav; Zografos, Odysseas; Gurunarayanan, Surya; Radu, Iuliana; Soree, Bart; Catthoor, Francky; Naeemi, Azad

被引用次数: 30

2017

Overcoming thermal noise in non-volatile spin wave logic

SCIENTIFIC REPORTS

Dutta, Sourav; Nikonov, Dmitri E.; Manipatruni, Sasikanth; Young, Ian A.; Naeemi, Azad

被引用次数: 6

2017

Non-volatile spin wave majority gate at the nanoscale

AIP ADVANCES

Zografos, O.; Dutta, S.; Manfrini, M.; Vaysset, A.; Soree, B.; Naeemi, A.; Raghavan, P.; Lauwereins, R.; Radu, I. P.

被引用次数: 29

2017

Non-monotonic probability of thermal reversal in thin-film biaxial nanomagnets with small energy barriers

AIP ADVANCES

Kani, N.; Naeemi, A.; Rakheja, S.

被引用次数: 2

2017

Analytical models for coupling reliability in identical two-magnet systems during slow reversals

JOURNAL OF APPLIED PHYSICS

Kani, Nickvash; Naeemi, Azad

被引用次数: 7

2017

Ultra-High Mobility in Dielectrically Pinned CVD Graphene

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Nashed, Ramy; Pan, Chenyun; Brenner, Kevin; Naeemi, Azad

被引用次数: 7

2016

A Proposal for Energy-Efficient Cellular Neural Network Based on Spintronic Devices

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Pan, Chenyun; Naeemi, Azad

被引用次数: 33

2016

Smart Detector Cell: A Scalable All-Spin Circuit for Low Power Non-Boolean Pattern Recognition

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Aghasi, Hamidreza; Iraei, Rouhollah Mousavi; Naeemi, Azad; Afshari, Ehsan

被引用次数: 6

2016

Scaling Limits on All-Spin Logic

IEEE TRANSACTIONS ON MAGNETICS

Chang, Sou-Chi; Kani, Nickvash; Manipatruni, Sasikanth; Nikonov, Dmitri E.; Young, Ian A.; Naeemi, Azad

被引用次数: 5

2016

A Model Study of an Error-Free Magnetization Reversal Through Dipolar Coupling in a Two-Magnet System

IEEE TRANSACTIONS ON MAGNETICS

Kani, Nickvash; Chang, Sou-Chi; Dutta, Sourav; Naeemi, Azad

被引用次数: 15

2016

Impact of On-Chip Interconnect on the Performance of 3-D Integrated Circuits With Through-Silicon Vias: Part II

IEEE TRANSACTIONS ON ELECTRON DEVICES

Zhang, Xuchen; Kumar, Vachan; Oh, Hanju; Zheng, Li; May, Gary S.; Naeemi, Azad; Bakir, Muhannad S.

被引用次数: 7

2016

Impact of On-Chip Interconnect on the Performance of 3-D Integrated Circuits With Through Silicon Vias: Part I

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kumar, Vachan; Oh, Hanju; Zhang, Xuchen; Zheng, Li; Bakir, Muhannad S.; Naeemi, Azad

被引用次数: 6

2016

A Probability-Density Function Approach to Capture the Stochastic Dynamics of the Nanomagnet and Impact on Circuit Performance

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kani, Nickvash; Rakheja, Shaloo; Naeemi, Azad

被引用次数: 8

2016

Interconnect Design and Benchmarking for Charge-Based Beyond-CMOS Device Proposals

IEEE ELECTRON DEVICE LETTERS

Pan, Chenyun; Naeemi, Azad

被引用次数: 8

2016

System-Level Variation Analysis for Interconnection Networks at Sub-10-nm Technology Nodes Using Multiple Patterning Techniques

IEEE TRANSACTIONS ON ELECTRON DEVICES

Pan, Chenyun; Baert, Rogier; Ciofi, Ivan; Tokei, Zsolt; Naeemi, Azad

被引用次数: 8

2015

Technology/Circuit/System Co-Optimization and Benchmarking for Multilayer Graphene Interconnects at Sub-10-nm Technology Node

IEEE TRANSACTIONS ON ELECTRON DEVICES

Pan, Chenyun; Raghavan, Praveen; Ceyhan, Ahmet; Catthoor, Francky; Tokei, Zsolt; Naeemi, Azad

被引用次数: 22

2015

Evaluating Chip-Level Impact of Cu/Low-k Performance Degradation on Circuit Performance at Future Technology Nodes

IEEE TRANSACTIONS ON ELECTRON DEVICES

Ceyhan, Ahmet; Jung, Moongon; Panth, Shreepad; Lim, Sung Kyu; Naeemi, Azad

被引用次数: 13

2015

Technology/System Codesign and Benchmarking for Lateral and Vertical GAA Nanowire FETs at 5-nm Technology Node

IEEE TRANSACTIONS ON ELECTRON DEVICES

Pan, Chenyun; Raghavan, Praveen; Yakimets, Dmitry; Debacker, Peter; Catthoor, Francky; Collaert, Nadine; Tokei, Zsolt; Verkest, Diederik; Thean, Aaron Voon-Yew; Naeemi, Azad

被引用次数: 35

2015

Compact Physical Model for Crosstalk in Spin-Wave Interconnects

IEEE TRANSACTIONS ON ELECTRON DEVICES

Dutta, Sourav; Nikonov, Dmitri E.; Manipatruni, Sasikanth; Young, Ian A.; Naeemi, Azad

被引用次数: 2

2015

Adapting Interconnect Technology to Multigate Transistors for Optimum Performance

IEEE TRANSACTIONS ON ELECTRON DEVICES

Prasad, Divya; Ceyhan, Ahmet; Pan, Chenyun; Naeemi, Azad

被引用次数: 17

2015

Phase-dependent deterministic switching of magnetoelectric spin wave detector in the presence of thermal noise via compensation of demagnetization

APPLIED PHYSICS LETTERS

Dutta, Sourav; Nikonov, Dmitri E.; Manipatruni, Sasikanth; Young, Ian A.; Naeemi, Azad

被引用次数: 6

2015

Non-volatile Clocked Spin Wave Interconnect for Beyond-CMOS Nanomagnet Pipelines

SCIENTIFIC REPORTS

Dutta, Sourav; Chang, Sou-Chi; Kani, Nickvash; Nikonov, Dmitri E.; Manipatruni, Sasikanth; Young, Ian A.; Naeemi, Azad

被引用次数: 57

2015

A Paradigm Shift in Local Interconnect Technology Design in the Era of Nanoscale Multigate and Gate-All-Around Devices

IEEE ELECTRON DEVICE LETTERS

Pan, Chenyun; Naeemi, Azad

被引用次数: 48

2015

A Fast System-Level Design Methodology for Heterogeneous Multi-Core Processors Using Emerging Technologies

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Pan, Chenyun; Naeemi, Azad

被引用次数: 6

2015

Design and Analysis of Copper and Aluminum Interconnects for All-Spin Logic

IEEE TRANSACTIONS ON ELECTRON DEVICES

Chang, Sou-Chi; Iraei, Rouhollah Mousavi; Manipatruni, Sasikanth; Nikonov, Dmitri E.; Young, Ian A.; Naeemi, Azad

被引用次数: 29

2014

SPICE Circuit Modeling of PMA Spin Wave Bus Excited Using Magnetoelectric Effect

IEEE TRANSACTIONS ON MAGNETICS

Dutta, Sourav; Nikonov, Dmitri E.; Manipatruni, Sasikanth; Young, Ian A.; Naeemi, Azad

被引用次数: 22

2014

Airgap Interconnects: Modeling, Optimization, and Benchmarking for Backplane, PCB, and Interposer Applications

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Kumar, Vachan; Sharma, Rohit; Uzunlar, Erdal; Zheng, Li; Bashirullah, Rizwan; Kohl, Paul; Bakir, Muhannad S.; Naeemi, Azad

被引用次数: 24

2014

Design and Analysis of Si Interconnects for All-Spin Logic

IEEE TRANSACTIONS ON MAGNETICS

Chang, Sou-Chi; Manipatruni, Sasikanth; Nikonov, Dmitri E.; Young, Ian A.; Naeemi, Azad

被引用次数: 28

2014

Circuit Simulation of Magnetization Dynamics and Spin Transport

IEEE TRANSACTIONS ON ELECTRON DEVICES

Bonhomme, Phillip; Manipatruni, Sasikanth; Iraei, Rouhollah M.; Rakheja, Shaloo; Chang, Sou-Chi; Nikonov, Dmitri E.; Young, Ian A.; Naeemi, Azad

被引用次数: 36

2014

A Proposal for a Novel Hybrid Interconnect Technology for the End of Roadmap

IEEE ELECTRON DEVICE LETTERS

Pan, Chenyun; Naeemi, Azad

被引用次数: 52

2014

Roles of Doping, Temperature, and Electric Field on Spin Transport Through Semiconducting Channels in Spin Valves

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Rakheja, Shaloo; Naeemi, Azad

被引用次数: 8

2013

Cu Interconnect Limitations and Opportunities for SWNT Interconnects at the End of the Roadmap

IEEE TRANSACTIONS ON ELECTRON DEVICES

Ceyhan, Ahmet; Naeemi, Azad

被引用次数: 69

2013

Impact of Dimensional Scaling and Size Effects on Spin Transport in Copper and Aluminum Interconnects

IEEE TRANSACTIONS ON ELECTRON DEVICES

Rakheja, Shaloo; Chang, Sou-Chi; Naeemi, Azad

被引用次数: 33

2013

Cu/Low-k Interconnect Technology Design and Benchmarking for Future Technology Nodes

IEEE TRANSACTIONS ON ELECTRON DEVICES

Ceyhan, Ahmet; Naeemi, Azad

被引用次数: 8

2013

Hydrogenation of Graphene Nanoribbon Edges: Improvement in Carrier Transport

IEEE ELECTRON DEVICE LETTERS

Zheng, Peng; Bryan, Sarah E.; Yang, Yinxiao; Murali, Raghu; Naeemi, Azad; Meindl, James D.

被引用次数: 11

2013

Evaluation of the Potential Performance of Graphene Nanoribbons as On-Chip Interconnects

PROCEEDINGS OF THE IEEE

Rakheja, Shaloo; Kumar, Vachan; Naeemi, Azad

被引用次数: 115

2013

Graphene Nanoribbon Spin Interconnects for Nonlocal Spin-Torque Circuits: Comparison of Performance and Energy Per Bit With CMOS Interconnects

IEEE TRANSACTIONS ON ELECTRON DEVICES

Rakheja, Shaloo; Naeemi, Azad

被引用次数: 21

2012

Performance and Energy-per-Bit Modeling of Multilayer Graphene Nanoribbon Conductors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kumar, Vachan; Rakheja, Shaloo; Naeemi, Azad

被引用次数: 81

2012

Power Delivery for 3-D Chip Stacks: Physical Modeling and Design Implication

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Huang, Gang; Bakir, Muhannad S.; Naeemi, Azad; Meindl, James D.

被引用次数: 18

2012

Ultralow-Power Single-Wall Carbon Nanotube Interconnects for Subthreshold Circuits

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Jamal, Omer; Naeemi, Azad

被引用次数: 21

2011

Modeling Interconnects for Post-CMOS Devices and Comparison With Copper Interconnects

IEEE TRANSACTIONS ON ELECTRON DEVICES

Rakheja, Shaloo; Naeemi, Azad

被引用次数: 12

2011

Interconnect Network Analysis of Many-Core Chips

IEEE TRANSACTIONS ON ELECTRON DEVICES

Balakrishnan, Anant; Naeemi, Azad

被引用次数: 8

2011

Interconnects for Novel State Variables: Performance Modeling and Device and Circuit Implications

IEEE TRANSACTIONS ON ELECTRON DEVICES

Rakheja, Shaloo; Naeemi, Azad

被引用次数: 12

2010

Optimal Global Interconnects for Networks-on-Chip in Many-Core Architectures

IEEE ELECTRON DEVICE LETTERS

Balakrishnan, Anant; Naeemi, Azad

被引用次数: 7

2010

Compact Physics-Based Circuit Models for Graphene Nanoribbon Interconnects

IEEE TRANSACTIONS ON ELECTRON DEVICES

Naeemi, Azad; Meindl, James D.

被引用次数: 202

2009

Carbon Nanotube Interconnects

ANNUAL REVIEW OF MATERIALS RESEARCH

Naeemi, Azad; Meindl, James D.

被引用次数: 101

2009

Electron transport modeling for junctions of zigzag and armchair graphene nanoribbons (GNRs)

IEEE ELECTRON DEVICE LETTERS

Naeemi, Azad; Meindl, James D.

被引用次数: 18

2008

Design and performance modeling for single-walled carbon nanotubes as local, semiglobal, and global interconnects in gigascale integrated systems

IEEE TRANSACTIONS ON ELECTRON DEVICES

Naeemi, Azad; Meindl, James D.

被引用次数: 198

2007

Conductance modeling for graphene nanoribbon (GNR) interconnects

IEEE ELECTRON DEVICE LETTERS

Naeemi, Azad; Meindl, James D.

被引用次数: 231

2007

Compact physical models for multiwall carbon-nanotube interconnects

IEEE ELECTRON DEVICE LETTERS

Naeemi, Azad; Meindl, James D.

被引用次数: 284

2006

Performance comparison between carbon nanotube and copper interconnects for gigascale integration (GSI)

IEEE ELECTRON DEVICE LETTERS

Naeemi, A; Sarvari, R; Meindl, JD

被引用次数: 250

2005

Impact of electron-phonon scattering on the performance of carbon nanotube interconnects for GSI

IEEE ELECTRON DEVICE LETTERS

Naeemi, A; Meindl, JD

被引用次数: 49

2005

Monolayer metallic nanotube interconnects: Promising candidates for short local interconnects

IEEE ELECTRON DEVICE LETTERS

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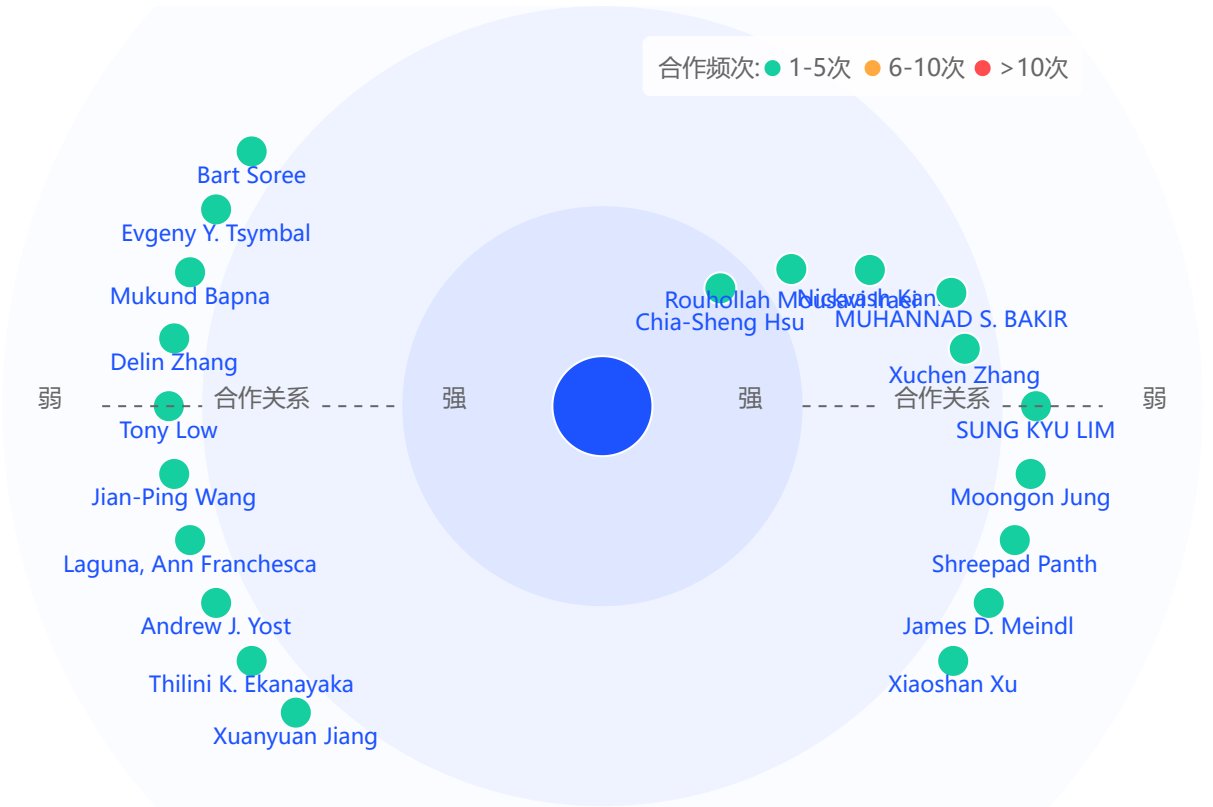
2004

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IEEE TRANSACTIONS ON ELECTRON DEVICES

Naeemi, A; Davis, JA; Meindl, JD	2004
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